

New

Home

Learn

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Discover

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie Agents

Alerts

Query History

SQL Warehouses

Data Engineering

Runs

Customer_Support_Pipeline

File Edit View Run Help Python Last edit was 7 hours ago

Run all Serverless Schedule Share

catalog schema
hive_metastore default

```
12:46 PM (<1s)
# Use workspace catalog
CATALOG = "workspace"
SCHEMA = "default"

# Fully-qualified table references
TBL_DAY1 = f"{CATALOG}.{SCHEMA}.day1_tickets"
TBL_DAY2 = f"{CATALOG}.{SCHEMA}.day2_tickets"
TBL_PROFILES = f"{CATALOG}.{SCHEMA}.agent_profiles"

print(f"Target Paths Initialized:\n {TBL_DAY1}\n {TBL_DAY2}\n {TBL_PROFILES}")

Target Paths Initialized:
workspace.default.day1_tickets
workspace.default.day2_tickets
workspace.default.agent_profiles
```

```
12:50 PM (21s)
from pyspark.sql import functions as F
from pyspark.sql.types import StructType, StructField, StringType, IntegerType
```


 New

 Home

 Learn

 Workspace

 Recents

 Catalog

 Jobs & Pipelines

 Compute

 Discover

 Marketplace

SQL

 SQL Editor

 Queries

 Dashboards

 Genie Agents

 Alerts

 Query History

 SQL Warehouses

Data Engineering

 Runs

 Data Ingestion

 Visual Data Prep

 Customer_Support_Pipeline x +

File Edit View Run Help Python  Last edit was now

  Run all  Serverless

catalog  schema 
hive_metastore default

```
12:50 PM (1s) 4 Python 

#Silver Layer
import re

def parse_resolution_time(time_str):
    if not time_str: return None
    m = re.match(r"(\d+)\s*h\s*(\d+)\s*m\s*(\d+)\s*s", time_str.strip(), re.IGNORECASE)
    if not m: return None
    h, mins, secs = int(m.group(1)), int(m.group(2)), int(m.group(3))
    total = h * 60 + mins
    if secs >= 30: total += 1
    return total

parse_time_udf = F.udf(parse_resolution_time, IntegerType())

silver_day1_times = clean_day1.withColumn("status_clean", F.upper(F.trim(F.col("status")))).withColumn("resolved_minutes", parse_time_udf(F.col("resolution_time"))).filter(F.col("resolved_minutes").isNotNull())
silver_day2_times = clean_day2.withColumn("status_clean", F.upper(F.trim(F.col("status")))).withColumn("resolved_minutes", parse_time_udf(F.col("resolution_time"))).filter(F.col("resolved_minutes").isNotNull())

> silver_day1_times: pyspark.sql.connect.dataframe.DataFrame = [ticket_id: string, agent_id: string ... 6 more fields]
> silver_day2_times: pyspark.sql.connect.dataframe.DataFrame = [ticket_id: string, agent_id: string ... 6 more fields]
```

ritikchanana64@gmail.com

workspace
Workspace ID: 7474651863372568 

OpenSharing ID: aws:u...e5213fc3 

Settings

Privacy policy

Send feedback

Log out

+ New

Home

Learn

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Discover

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie Agents

Alerts

Query History

SQL Warehouses

Customer_Support_Pipeline

File Edit View Run Help Python Last edit was now

catalog schema
hive_metastore default

1250 PM (1s) 4 Python
> silver_day1_times: pyspark.sql.connect.dataframe.DataFrame = [ticket_id: string, agent_id: string ... 6 more fields]
> silver_day2_times: pyspark.sql.connect.dataframe.DataFrame = [ticket_id: string, agent_id: string ... 6 more fields]

1251 PM (1s) 5 Python
IN_SCOPE_TLS = {f"TL{str(i).zfill(2)}" for i in range(1, 9)}

def process_filters(tickets_df, profiles_df):
 enriched = tickets_df.join(F.broadcast(profiles_df), on="agent_id", how="inner")
 scoped = enriched.filter(F.col("team_lead_id").isin(IN_SCOPE_TLS))
 # Keep strictly higher than 15 minutes and matching 'RESOLVED' status
 return scoped.filter((F.col("status_clean") == "RESOLVED") & (F.col("resolved_minutes") > 15))

silver_day1_success = process_filters(silver_day1_times, bronze_profiles)
silver_day2_success = process_filters(silver_day2_times, bronze_profiles)

> silver_day1_success: pyspark.sql.connect.dataframe.DataFrame = [agent_id: string, ticket_id: string ... 9 more fields]
> silver_day2_success: pyspark.sql.connect.dataframe.DataFrame = [agent_id: string, ticket_id: string ... 9 more fields]

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Settings

Privacy policy

Send feedback

Log out

+

New

Home

Learn

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Discover

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie Agents

Alerts

Query History

SQL Warehouses

Data Engineering

Runs

Data Ingestion

Visual Data Prep

AI/ML

Playground

Customer_Support_Pipeline

File Edit View Run Help Python ☆ Last edit was 1 minute ago

catalog

hive_metastore

schema

default

12:51 PM (28s)

6

Python

```
# Identify successful agents from Day 1 to drop from Day 2 data execution
day1_successful_agents = silver_day1_success.select("agent_id").distinct()
silver_day2_carryover = silver_day2_success.join(day1_successful_agents, on="agent_id", how="left_anti")

# Combine results
gold_combined = silver_day1_success.union(silver_day2_carryover)

print("=== Q1 - Resolution Rates by Team Lead ===")
gold_q1 = gold_combined.groupBy("team_lead_id").agg(F.count("ticket_id").alias("total_resolved_tickets")).orderBy(F.col("total_resolved_tickets").desc())
display(gold_q1)

print("\n=== Q2 - Per-Agent Summary Matrix ===")
gold_q2 = gold_combined.groupBy("agent_id", "agent_name", "Day").agg(F.count("ticket_id").alias("resolved_count")).orderBy("agent_id")
display(gold_q2)
```

See performance (2)

✓ Checked for improvements

day1_successful_agents: pyspark.sql.connect.dataframe.DataFrame = [agent_id: string]

gold_combined: pyspark.sql.connect.dataframe.DataFrame = [agent_id: string, ticket_id: string ... 9 more fields]

gold_q1: pyspark.sql.connect.dataframe.DataFrame = [team_lead_id: string, total_resolved_tickets: long]

gold_q2: pyspark.sql.connect.dataframe.DataFrame = [agent_id: string, agent_name: string ... 2 more fields]

silver_day2_carryover: pyspark.sql.connect.dataframe.DataFrame = [agent_id: string, ticket_id: string ... 9 more fields]

=== Q1 - Resolution Rates by Team Lead ===

Table

+

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	team_lead_id	total_resolved_tickets
1	TL01	9

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Settings

Privacy policy

Send feedback

Log out

	team_lead_id	total_resolved_tickets
1	TL01	9

New

Home

Learn

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Discover

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie Agents

Alerts

Query History

SQL Warehouses

Data Engineering

Runs

Data Ingestion

Visual Data Prep

AI/ML

Playground

Customer_Support_Pipeline

File Edit View Run Help Python Last edit was 2 minutes ago

catalog

schema

hive_metastore

default

=== Q1 - Resolution Rates by Team Lead ===

Table

	team_lead_id	total_resolved_tickets
1	TL01	9
2	TL02	2

2 rows | 23.098s runtime

Refreshed 7 hours ago

=== Q2 - Per-Agent Summary Matrix ===

Table

	agent_id	agent_name	Day	resolved_count
1	A001	Agent_A001	1	2
2	A002	Agent_A002	1	2
3	A003	Agent_A003	1	2
4	A004	Agent_A004	1	2
5	A005	Agent_A005	1	1
6	A006	Agent_A006	1	2

6 rows | 23.098s runtime

Refreshed 7 hours ago

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Settings

Privacy policy

Send feedback

Log out

databricks

Free Edition

New

Home

Learn

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Discover

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie Agents

Alerts

Query History

SQL Warehouses

Data Engineering

Runs

Data ingestion

Visual Data Prep

AI/ML

Customer_Support_Pipeline

File Edit View Run Help Python ☆ Last edit was 2 minutes ago

catalog

hive_metastore

schema

default

6 rows | 28.098s runtime

7

```
# Validate Team Lead distribution
display(bronze_profiles.groupBy("team_lead_id").count().orderBy("team_lead_id"))

# Validate Day 1 ticket statuses
display(bronze_day1.groupBy("status").count())

# Validate Day 2 ticket statuses
display(bronze_day2.groupBy("status").count())

# Final Gold Report 1
display(gold_q1)

# Final Gold Report 2
display(gold_q2.orderBy("Day", "agent_id"))

# Record counts
print("Profiles:", bronze_profiles.count())
print("Day1:", bronze_day1.count())
print("Day2:", bronze_day2.count())
print("Gold TL:", gold_q1.count())
print("Gold Agent:", gold_q2.count())
```

Table +

Run all

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Settings

Privacy policy

Send feedback

Log out

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https://dbc-d6915243-cb33.cloud.databricks.com/recents?o=747465186337...

New

Home

Learn

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Discover

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie Agents

Alerts

Query History

SQL Warehouses

Data Engineering

Runs

Data Ingestion

Visual Data Prep

AI/ML

Playground

Customer_Support_Pipeline

catalog schema

hive_metastore default

	team_lead_id	count
1	TL01	5
2	TL02	5
3	TL03	5
4	TL04	5
5	TL05	5
6	TL06	5
7	TL07	5
8	TL08	5
9	TL09	1
10	TL12	1

10 rows

	status	count
1	Resolved	16
2	Open	2
3	Pending	2

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Workspace ID: 7474651863372568

OpenSharing ID: awsu...e5213fc3

Settings

Privacy policy

Send feedback

Log out

databricks

FREE EDITION

New

Home

Learn

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Discover

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie Agents

Alerts

Query History

SQL Warehouses

Data Engineering

Runs

Data Ingestion

Visual Data Prep

AI/ML

Playground

Customer_Support_Pipeline

File Edit View Run Help Python ☆ Last edit was 3 minutes ago

catalog

hive_metastore

schema

default

2 rows

Table

	agent_id	agent_name	Day	resolved_count
1	A001	Agent_A001	1	2
2	A002	Agent_A002	1	2
3	A003	Agent_A003	1	2
4	A004	Agent_A004	1	2
5	A005	Agent_A005	1	1
6	A006	Agent_A006	1	2

6 rows

Profiles: 42

Day1: 20

Day2: 8

Gold TL: 2

Gold Agent: 6

Run all

Serverless

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Workspace ID: 7474651863372568

OpenSharing ID: awsu...e5213fc3

Settings

Privacy policy

Send feedback

Log out

	agent_id	agent_name	Day	resolved_count
1	A001	Agent_A001	1	2
2	A002	Agent_A002	1	2
3	A003	Agent_A003	1	2
4	A004	Agent_A004	1	2
5	A005	Agent_A005	1	1
6	A006	Agent_A006	1	2